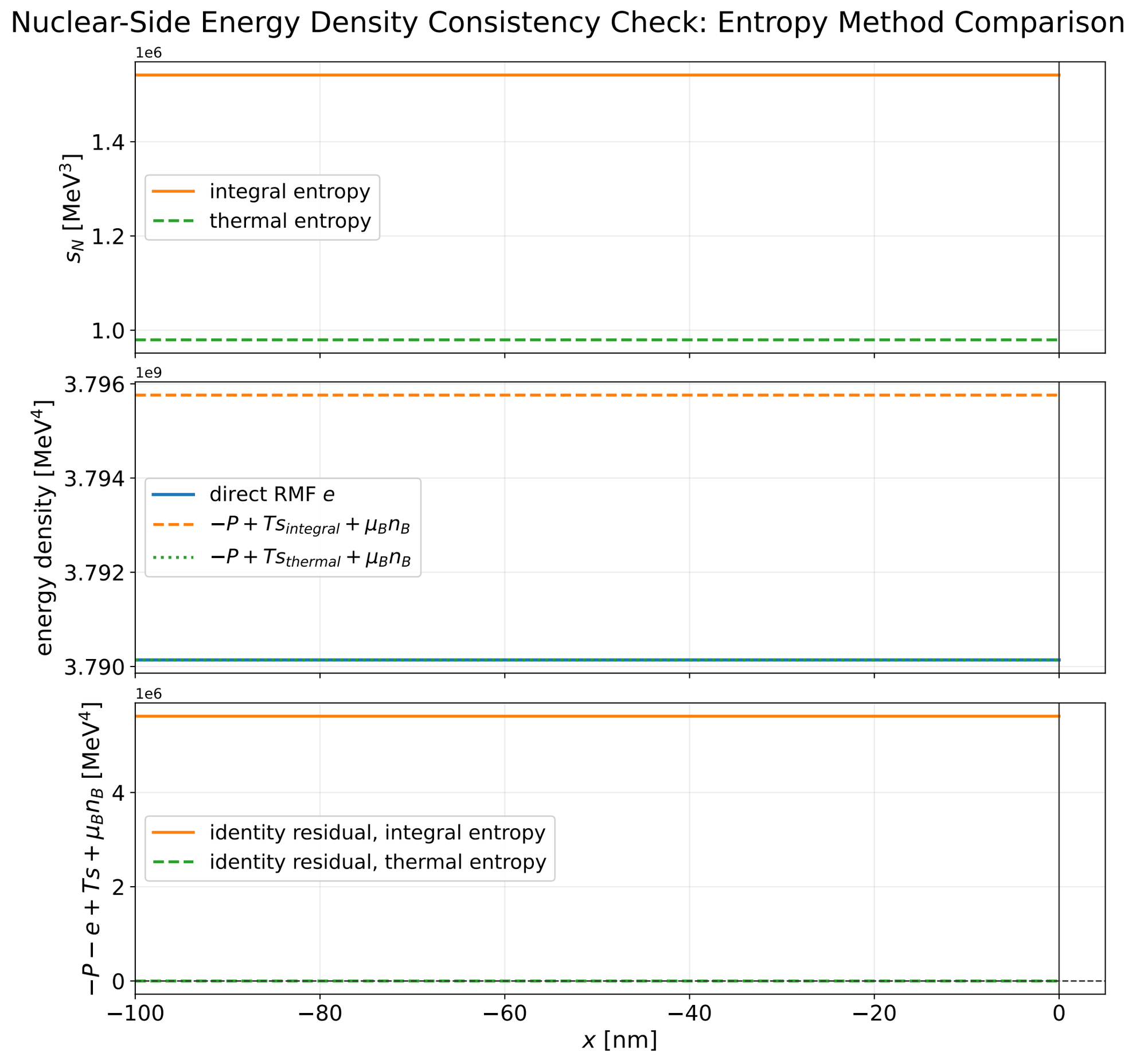
Nuclear-Side Energy Density Consistency Check: Entropy Method Comparison
1e6
1.4
1.2
1.0
s_N [MeV^3]
integral entropy
thermal entropy
1e9
3.796
3.794
3.792
3.790
energy density [MeV^4]
direct RMF e
$-P + Ts_{integral} + \mu_B n_B$
$-P + Ts_{thermal} + \mu_B n_B$
1e6
4
2
0
$-P - e + Ts + \mu_B n_B$ [MeV^4]
identity residual, integral entropy
identity residual, thermal entropy
−100
−80
−60
−40
−20
0
x [nm]